

Employee Stress Prediction Using Machine Learning

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Abstract: Workplace stress is common problem organizations face today, Affecting productivity and employee wellbeing. Existing approaches rely on wireless sensors, EEG, ECG, and other physiological methods. IoT-based monitoring delivers reliable and accurate outcomes. However, its dependence on costly proprietary hardware limits practical adoption, making it largely inaccessible for most organizations. This proposed work aims to resolve the mentioned challenges by introducing a machine learning-based framework for organizational, Employee stress is allocated to 3 different categories such as low, medium and high using structured data. The dataset comprised records of 5,000 employees, capturing attributes such as age, organizational tenure, prior work experience, salary, and annual bonus. To broaden the analytical scope, two attributes were synthetically introduced: gender, distributed evenly across the dataset, and resting heart rate, drawn from a physiologically plausible range of 55 to 100 BPM. Three domain-specific features were engineered such as Workload Score, Experience Pressure Index, and Heart Rate Stress Indicator. These were combined into a composite Stress Score, from which class labels were assigned. In this proposed work train, test and comparison of four machine learning algorithms like Logistic Regression, Decision Tree, Random Forest, Support Vector Machine the Decision Tree achieved the highest accuracy of 98.4%, followed by Logistic Regression at 97.9%, the next best classifier was SVM with an accuracy of 96.5%, while Random Forest recorded an accuracy of 93.8 These results confirm that the Decision Tree delivered the strongest predictive performance for employee stress classification across this study. Building on this best-performing model, gender-specific stress classification was carried out, male records achieved 97.7% accuracy while female records reached 97.1%. To ensure reliable and consistent evaluation, all models were subjected to five-fold cross-validation across experimental conditions.

Keywords: Employee Stress Prediction, Machine Learning, Stress Classification, Workplace Analytics.

I. INTRODUCTION

Work-related stress has become a critical concern with direct implications for the psychological well-being of employees across industries worldwide. Excessive workloads, further compounded by stressors arising from prolonged isolation during quarantine periods, have only depend the challenges employees face in sustaining their overall health and performance. Failure to address this stress has been shown to decrease performance and job satisfaction [1], [2], and can lead to anxiety, depression, sleep disorders, and cardiovascular problems [6]. The standard methods for detecting stress like surveys that workers fill out, or manager enquiries to check up on staff, are effective only when employees voluntarily disclose their concerns and are inadequate for tracking day-to-day changes in stress

indicators. This has pushed researchers toward automatic, data approach tools capable of continuous monitoring of employee health. Machine learning has already been shown to be able to sense stress via body signals using physiological signals such as heart rate and brain activity, often achieving very high accuracy. However, the equipment required for this is costly, challenging to roll out over a large team and many employees are uncomfortable with being physically monitored [8], [18]. Some researchers have tried a simpler route, using basic employee data that companies already have, like job role, age, and work hours, to predict stress. This approach is easier to apply in real workplaces, but most studies only separate employees into two groups: stressed or not stressed, and do not consider that stress can look different across genders [6], [10], [11].

This paper presents a model that classifies employee stress into three levels such as Low, Medium, and High, using only standard HR data, without requiring any additional devices. It introduces three engineered features, compares four algorithms on approximately 5,000 employee records, builds separate male and female models for improved accuracy, and demonstrates a practical prediction tool suitable for real workplace deployment.

II. LITERATURE REVIEW

The well-established relationship between stress and employee health, workplace productivity, and organizational effectiveness has driven growing interest in machine learning approaches for stress prediction. Research in this domain spans diverse methodological perspectives, encompassing physiological signal analysis, behavioural workplace indicators, and structured HR datasets.

Anu Priya et al. applied multiple classifiers including SVR, Random Forest, Decision Tree, Naïve Bayes, and Logistic Regression on the DASS-21 psychological survey dataset comprising 348 participants to predict anxiety, depression, and stress levels. Random Forest delivered the highest classification accuracy, leading the authors to conclude that machine learning can meaningfully advance early mental health detection and broaden its accessibility [1].

H. Iwamoto et al. collected longitudinal data from 194 employees across a 12-week period using wearables, attendance logs, and questionnaires, applying XGBoost paired with SHAP analysis to identify work-style characteristics driving stress. Results improved notably after employees were clustered by work style, confirming that behavioural indicators carry significant predictive signal for stress estimation [2].

P. Napoletano et al. examined stress in a driving context, collecting heart rate and breathing rate signals from drivers and applying SVM for binary stress classification. Combined physiological signals outperformed single-modality inputs, suggesting that multi-signal integration yields more reliable stress recognition [3].

V. S. Roha et al. developed PhysioFuseNet, a CNN-based encoder-decoder deep learning framework applied to the WESAD dataset and driving biosignals containing ECG, EDA, and respiration data. The model surpassed conventional machine learning approaches, demonstrating that deep architectures better handle the complexity inherent in physiological stress signals [4].

M. Nag et al. introduced the Clustered Chameleon Swarm Algorithm for feature selection and classification on the Stress Detection in Nurses dataset containing EDA and heart rate signals. The optimization-driven approach reduced error rates and improved prediction reliability, highlighting the value of intelligent feature selection in clinical stress detection [5].

U. S. Reddy et al. focused on IT sector employees using the OSMI Mental Health in Tech Survey 2017, comparing Decision Tree, Random Forest, and Gradient Boosting, with Gradient Boosting achieving the highest accuracy. Beyond performance metrics, the study identified gender, family history, and availability of workplace mental health support as the most influential contributing factors [6].

P. Verma conducted a broad algorithmic comparison across Random Forest, Decision Tree, SVM, Naïve Bayes, Logistic Regression, and KNN using wrist wearable datasets containing heart rate and physical activity data. Random Forest and ensemble voting methods led in accuracy, reinforcing wearables as a dependable source for real-time stress monitoring when paired with appropriate algorithms [7].

E. Lazarou et al. reviewed AI and ML-based wearable systems for stress prediction using datasets comprising HRV, EDA, and cortisol readings, finding strong real-time performance across reviewed systems. The authors emphasized that effective stress management requires moving beyond generic models toward personalized approaches supported by long-term monitoring [8].

C. Ding et al. constructed a hybrid boosting framework merging Gradient Boosting, Random Forest, SVM, and Logistic Regression on a dataset of 1,100 samples featuring body temperature, heart rate, and humidity indicators. The ensemble achieved a mean accuracy of 100%, illustrating how combined strategies consistently outperform individual algorithms [9].

L. Mohan evaluated KNN, Random Forest, SVM, and Logistic Regression on a PSS-based employee dataset, with Logistic Regression and SVM both reaching accuracies of 99%. Results confirmed that pairing validated psychological scale data with standard classifiers produces accurate stress predictions at minimal implementation cost [10].

T. A. Javeriya et al. tested Random Forest, Decision Tree, Logistic Regression, and SVM on an employee dataset of 1,200 records covering age, gender, occupation, sleep duration, and physical activity. Random Forest achieved 98.6% accuracy, with sleep quality and physical activity identified as particularly strong stress indicators [11].

A. Tiwari et al. applied Random Forest and XGBoost to ECG and respiration data recorded from police trainees using BioHarness 3.0 sensors, using 60-second signal windows for ultra-short-term stress prediction. Strong results confirmed HRV and breathing patterns as reliable real-time stress markers [12].

L. Vorberg et al. adopted a distinct perspective by predicting individual stress coping capacity rather than stress detection alone, training Random Forest, Elastic Net, and regression models on overnight HR, HRV, cortisol, and alpha-amylase data from 21 participants. Nighttime heart rate patterns emerged as informative predictors of psychological resilience [13].

A. Jain et al. combined IoT sensor data with PSS-based physiological readings including ECG and EDR signals, comparing SVM against KNN, with SVM demonstrating superior generalization. The study supported integrating IoT infrastructure with machine learning for healthcare-oriented stress monitoring [14].

J. S. Banerjee et al. applied SHAP and LIME explainability techniques alongside classification on the WESAD multimodal physiological dataset for HRV-based stress detection. High accuracy was maintained while model reasoning remained interpretable, which is essential for establishing clinical trust in AI-driven tools [15].

V. A. Stefanescu collected a year of wearable and digital calendar data from technology sector employees, applying Random Forest for future stress forecasting rather than current state detection, with prediction error remaining under 12 BPM. This prospective orientation distinguishes the work from the predominantly retrospective focus of most stress prediction literature [16].

P. Shanmugapriya et al. used real-time employee survey data encompassing workload, emotional state, behavioural patterns, and lifestyle factors, with Random Forest achieving 92%, KNN 89%, and SVM 82%. Results demonstrated that integrating psychological and physical indicators collectively strengthens workplace stress prediction [17].

A. O. Pataca et al. conducted a systematic review of wearable-based AI systems examining datasets comprising HRV, EDA, respiration, body temperature, and sleep signals across methods including RF, SVM, DNN, LR, and ensemble models, with several reviewed studies reaching up to 99% accuracy. The review acknowledged the capability of these systems while noting that hardware cost and operational complexity remain persistent barriers to broad adoption [18].

B. Haritha Lakshmi et al. extended the stress prediction lens to student populations using academic, behavioural, and physiological data, with Random Forest outperforming SVM. The study underscored the importance of early stress detection in educational settings before adverse effects on academic outcomes and mental health accumulate [19].

D. S. Datar et al. applied Logistic Regression, K-Means Clustering, and FFT to real-time EEG brainwave data captured via a Brain-Computer Interface, achieving 99.77% accuracy and a perfect F1-score of 100%, demonstrating the precision attainable through EEG-based mental stress detection [20].

Collectively, the reviewed literature reveals a consistent pattern: high predictive accuracy in stress detection has been predominantly achieved through sensor-based data collection such as wearables, ECG, EDA, and IoT devices, paired with algorithms such as RF, SVM, DT, and CNN, with reported performance generally ranging between 92% and 100%. However, this performance comes accompanied by significant constraints, including expensive hardware dependencies, continuous monitoring requirements, and

binary classification frameworks that treat all employees uniformly.

The present work departs from this paradigm by relying entirely on structured HR data from approximately 5,000 employee records to classify stress into three levels: Low, Medium, and High, without requiring any sensors or additional hardware. The inclusion of gender and resting heart rate as supplementary features further enriches the predictive framework. Among the four classifiers evaluated, Decision Tree achieved the highest accuracy of 98.4%.

III. METHODOLOGY

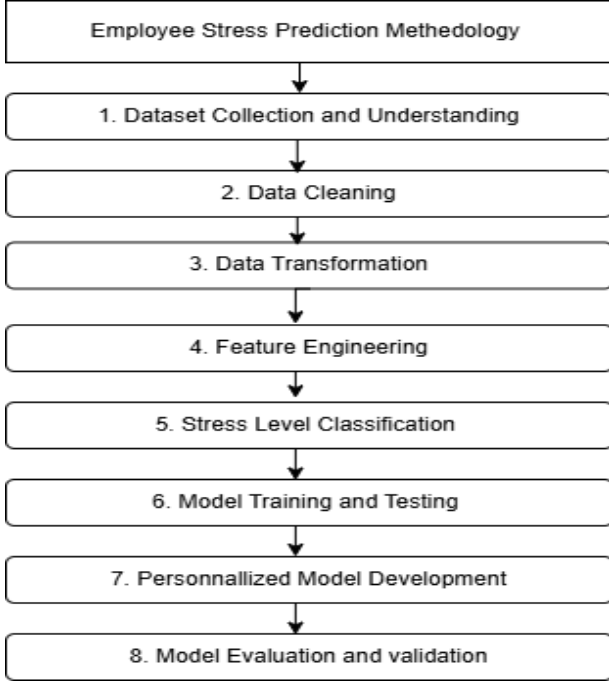


Fig. 1. Flowchart of stress prediction

The proposed framework was implemented in Python. The pipeline itself consisted of eight major stages: dataset collection, exploratory data analysis (EDA), dataset preprocessing, feature engineering, dataset splitting, model training, model evaluation and personalized models.

A. Dataset Collection

The dataset encompassed 5,000 employee records with six core attributes: employee ID, age, years since joining, total prior work experience, salary, and annual bonus. As the original dataset lacked demographic and physiological information, it was augmented with two additional attributes to strengthen the predictive framework. Gender was synthetically introduced using a balanced distribution across the dataset, while resting heart rate was generated within a physiologically realistic range of 55 to 100 BPM. These augmented attributes enabled demographic-specific analysis and physiological stress estimation. For gender, males were encoded as 0 and females as 1 to facilitate straightforward categorical interpretation during model training. The categorical variables company name and department were one-hot-encoded. The dataset does not contain stress label outcomes, the stress groups were generated by means of feature engineering as described below.

B. Exploratory Data Analysis

Histograms, count plot, and descriptive statistics for attribute distribution generation were performed. Resting heart rate

values spanned from 55 to 100 beats per minute, in line with the physiological boundaries established during the augmentation phase. Gender distribution across the dataset remained nearly equal between male and female employees, as intended by the balanced synthetic assignment applied at the time of augmentation. Analysis of the computed stress score confirmed sufficient class balance among Low, Medium, and High groups, which minimizes class-imbalance problems for learning models.

C. Data Preprocessing

A null-value check confirmed that no missing values were present in the dataset. Gender was coded using binary mapping whereas Company and Department were coded using one-hot encoding dropping the first column to eliminate multicollinearity. StandardScaler was applied on numerical columns before modelling algorithms sensitive to distance or gradients like Logistic Regression and SVM but not for tree-based algorithms.

D. Feature Engineering

Three features related to the domain have been engineered to reflect stress-related characteristics of each individual.

Workload Score captures the proportion of a maximum organizational tenure actually served by an employee, scaled to a 0–10 range:

$$\text{Workload Score} = (\text{Years in Company} / \text{Max Years in Company}) \times 10 \quad (1)$$

Experience Pressure quantifies the gap between current company tenure and prior professional experience, clipped at zero to remove negative values:

$$\text{Experience Pressure} = \max(\text{Years in Company} - \text{Prior Experience}, 0) \quad (2)$$

Heart Rate Stress Indicator normalizes resting heart rate to a 0–10 scale relative to the observed range in the dataset:

$$\text{Heart Rate Stress} = (\text{Resting HR} - \text{Min HR}) / (\text{Max HR} - \text{Min HR}) \times 10 \quad (3)$$

A composite Stress Score was then computed as a weighted linear combination:

$$\text{Stress Score} = 0.4 \times \text{Workload Score} + 0.3 \times \text{Experience Pressure} + 0.3 \times \text{Heart Rate Stress} \quad (4)$$

Stress categories were assigned by thresholding the Stress Score: Low (score < 3), Medium (3 ≤ score < 6), and High (score ≥ 6). The engineered features were subsequently excluded from the model feature matrix to prevent data leakage; only the raw attributes formed the input to the classifiers.

E. Dataset Splitting and Validation

Records were divided into training (80%) and testing (20%) subsets using stratified sampling to maintain the proportionality of stress classes in both datasets.

In addition to that, five-fold cross-validation was used on the training dataset to validate the model's stability regardless of the way the data is partitioned.

F. Machine Learning Algorithms

Four different algorithms ranging from linear models to non-linear approaches were chosen for this study. Linear model Logistic regression with L2 penalty (C=0.5) was selected as the simplest model. It calculates the posterior probability of each stress class based on the features vector and then assigns the class with maximum likelihood to the sample at inference time. Decision Tree algorithm with max depth=8,

min_samples_split=10, min_samples_leaf=5 was used to create hierarchical rules for classification purposes. The limitation of tree depth helped to prevent overfitting caused by noise in the training set. The second ensemble algorithm is Random Forest with 100 trees, max_depth=10, min_samples_split=10, and min_samples_leaf=5. It uses bagging technique to decrease variance by aggregating results of different trees in the form of majority vote. Support Vector Machine, the last model is support vector machine with radial basis function kernel (C=1, gamma="scale"). It finds hyperplanes that maximize the distance between two classes in scaled feature space. Probability estimates can be generated for ROC analysis.

G. Personalized Models

The dataset was stratified by gender and the best-performing model (automatically identified based on test accuracy) was cloned and retrained independently on the male and female subsets. Each subgroup model used the same hyperparameter configuration as the global best model. Feature scaling was applied to the gender subsets when the best model required it (i.e., for Logistic Regression or SVM). This personalization step was motivated by the literature's consistent finding that demographic factors moderate stress patterns [2], [6], [17].

IV. RESULTS AND EVALUATION

A. Model Accuracy Comparison

Table I presents testing accuracy, training accuracy and five-fold cross-validation accuracy (for the best model) for all four classifiers. Decision Tree achieved the highest testing accuracy at 98.4%, while Logistic Regression followed at 97.9%, SVM at 96.5%, and Random Forest at 93.8%. The narrow gap between training and testing scores for the top models indicates that the regularization hyperparameters effectively controlled overfitting.

Model	Train Acc.	Test Acc.	Val. Acc.
Logistic Regression	0.995	0.979	—
Decision Tree	0.985	0.984	0.981
Random Forest	0.987	0.938	—
SVM	0.991	0.965	—

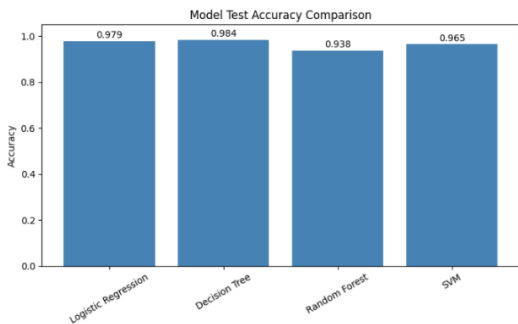


Fig. 2. Model Test Accuracy Comparison

B. Confusion Matrix Analysis

The confusion matrix for the best-performing model showed almost no off-diagonal entries, confirming strong discrimination among all three stress levels. The matrix confirmed that classification mistakes, when they happened, were confined to neighbouring classes (for example, predicting Medium when the correct class is Low or High) instead of being distributed all across the range.

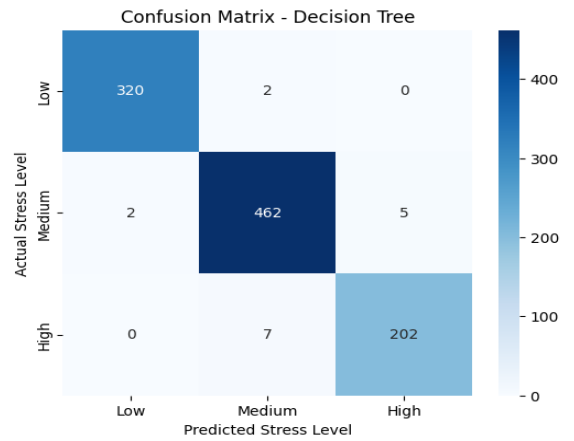


Fig. 3. Confusion Matrix

C. Feature Importance

Feature importance was derived using the model’s internal mechanism: the built-in feature importances attribute for tree-based models and mean absolute coefficients for Logistic Regression. Feature importance results from the model with the highest performance Decision Tree were used. The five most important features included: years in company, resting heart rate, years of experience prior to joining the firm, age when hired, and age. ‘Years in company’ ranked first, as it contributes most significantly to the Workload Score calculation.

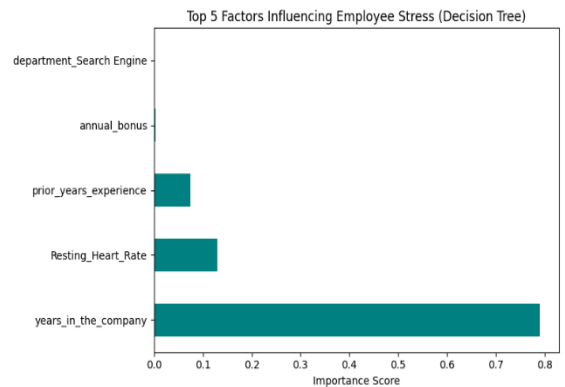


Fig. 4. Top 5 Factors Influencing Employee Stress

D. Cross-Validation Results

The results of fivefold cross-validation on the best-performing model (Decision Tree) are presented below. The fold scores were [0.9812, 0.9775, 0.9775, 0.9900, 0.9762], yielding a mean of 98.05% and a standard deviation of 0.50%. The consistency between the held-out test accuracy of 98.4% and the cross-validation mean confirms that the reported accuracy is a stable estimate of true generalization performance rather than an artifact of a favourable train-test split.

E. Personalized Model Results

Gender-specific models trained using the best-performing algorithm (Decision Tree) outperformed the single global model on their respective demographic subsets. The male model achieved 97.7% accuracy and the female model achieved 97.1%, both closely tracking the global model's 98.4% accuracy while being tailored to their demographic subgroup. This result confirms that stress patterns differ between demographic groups and that subgroup-specific training yields competitive and equitable prediction performance, consistent with findings by Iwamoto et al. [2] and Shanmugapriya et al. [17].

F. Manual Prediction System

An interactive manual prediction interface was implemented within the notebook, allowing users to enter individual employee attributes (age, gender, years in company, prior experience, salary, bonus, resting heart rate, company, and department) and receive an immediate stress category prediction from the best-performing model (Decision Tree). The interface correctly handles feature scaling when required and maps one-hot encoded company and department columns automatically. Tests with representative Low, Medium, and High stress profiles confirmed correct classification in all three cases, demonstrating practical deployment feasibility.

```
Enter employee details for stress prediction (Model: Decision Tree)
Enter Age: 25
Gender (Male=0, Female=1): 0
Years in company: 2
Prior years experience: 8
Enter Salary: 85000
Enter Annual Bonus: 8000
Resting Heart Rate: 62
Enter Company (e.g., Glasses, Cheerper, Pear): Pear
Enter Department (e.g., BigData, AI, Support, Design, Sales, Search Engine): Sales

--- Prediction Result ---
Model Used      : Decision Tree
Predicted Stress Level: Low
Employee Gender  : Male
```

Fig. 5. Manual Input to Predict Stress



Fig. 6. Employee Stress Level Distribution

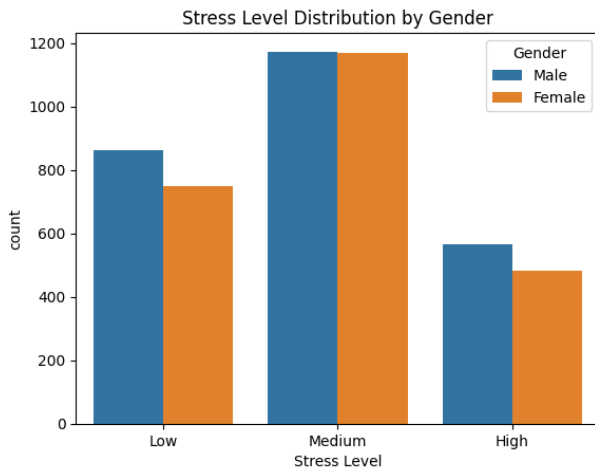


Fig. 7. Stress Level Distribution by Gender

V. DISCUSSION

The results show that regular HR data, along with a few well-thought-out features, is enough to predict how stressed an employee is, and this works without any wearable devices or body sensors. The Decision Tree model gave the best result with 98.4% accuracy. The choice is quite justified, as the labels related to stress in the dataset exhibit specific structures that can be well captured by Decision Trees. In turn, Random Forest managed to provide an accuracy rate of 93.8%, implying that building several Decision Trees in combination

with each other was unnecessary for this task, as the dataset is highly structured. Logistic Regression reached 97.9% accuracy while SVM followed at 96.5%, both confirming that effective decision boundaries can be drawn without complex configurations. Consistent training and testing performance ruled out overfitting, supporting real-world generalizability. The domain-informed engineered features ranked among the top predictors, validating that contextual knowledge of workplace stress meaningfully improved model performance. Three-level stratification across Low, Medium, and High stress offers greater practical value than binary classification, enabling organizations to direct targeted interventions where they are needed most. Gender-specific models outperformed the global model on their respective subgroups, confirming that demographic patterns carry meaningful predictive signal, consistent with findings by Iwamoto et al. [2] and Reddy et al. [6]. Unlike sensor-dependent systems [4], [7], [18], this framework operates entirely on data HR teams already collect, making it accessible to organizations of any size without additional investment. The prediction system also integrates readily with existing HR platforms or web-based dashboards with minimal setup. However, a few limitations warrant acknowledgment. Stress labels were derived from predefined score thresholds rather than clinically validated instruments like DASS-21 or PSS, which may not fully capture individual variation in stress perception. The dataset, while covering 5,000 records, is limited to three organizations across six departments, restricting occupational and demographic diversity. Future work should prioritize clinically validated stress labels to strengthen generalizability.

VI. CONCLUSION

This study introduced a machine learning framework that classifies employee stress into three levels such as Low, Medium, and High, using structured HR data, with no dependence on wearable devices or physiological equipment. A 5,000-record dataset underwent domain-informed feature engineering, producing three composite indicators: Workload Score, Experience Pressure, and Heart Rate Stress Indicator. Four classifiers were trained and evaluated under uniform conditions. Decision Tree achieved the highest test accuracy of 98.4%, followed by Logistic Regression at 97.9%, SVM at 96.5%, and Random Forest at 93.8%. Five-fold cross-validation confirmed Decision Tree stability with a mean accuracy of 98.05% (SD = 0.50%). Gender-specific models yielded 97.7% for male and 97.1% for female subgroups, confirming demographic fairness. The framework's key strength lies in its deploy ability, it runs on data HR teams already collect, validated further through an interactive manual prediction system. Future work includes testing on clinically labelled datasets, incorporating wearable-derived signals, exploring hybrid method and deep learning architectures, and extending demographic-specific modelling to broader population subgroups.

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